

# Amar Dhillon

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## EDUCATION

**University of Maryland, College Park**  
*B.S. in Computer Science, Minor in Statistics; GPA: 3.5*

College Park, MD  
*Expected May 2028*

## CERTIFICATIONS

**Amazon Web Services (AWS) Certified Developer - Associate**  
*Amazon Web Services*

Nov 2024 – Nov 2027  
*Digital Badge*

## EXPERIENCE

**Software Development Engineer Intern**  
*Amazon Web Services*

Sep 2026 – Dec 2026  
*Redmond, WA*

- Writing Java logic that sets the upper and lower bounds on EC2 capacity that Amazon Redshift reserves for operations like patching, balancing availability against cost, and backtesting it

**Software Engineering Intern**  
*Capital One*

Jun 2026 – Aug 2026  
*Richmond, VA*

- Rebuilt the legacy tool developers use to create test-data workflows as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation, and **90%** of 30+ surveyed developers preferred it
- Cut AI workflow generation time **65%** to under 15s by replacing a Claude skill's open-ended search with focused sub-skills that call Python tools for workflow search and validation, guided by CloudWatch and OpenSearch latency logs
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a category tree that the Claude skill searches first, with Redis caching full workflows, so lookups rarely touch a database gated by permissions and rate limits

**Software Engineer (Contract)**  
*UMD App Development Contracting (Client: Amazon Leo)*

Jan 2026 – Present  
*College Park, MD*

- Co-built a research tool that scores and aggregates topics so Amazon Leo's public-policy staff can see what's popular, running in dev on AWS Fargate with GitHub Actions CI/CD and AWS CDK infrastructure-as-code ready for production
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Automated retraining for a production Amazon Bedrock classifier that rates research documents by importance: on an EventBridge schedule a larger LLM grades its ratings, and fine-tuning runs only when the grades shift past a threshold

**Machine Learning Research Intern**  
*UMD Institute for Advanced Computer Studies (UMIACS)*

Dec 2025 – Present  
*College Park, MD*

- For a PhD-led robotics project aimed at ICLR, built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D for a vision-language model, then tuned prompts so it favors grasps that avoid hitting the table

**Co-Founder, Software Engineer**  
*TerpLabs*

Jun 2025 – Present  
*College Park, MD*

- Co-founded a 110+ member student product studio that has shipped 5 live products
- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency
- Building a RAG meal-suggestion feature for the official UMD app, in which GPT-4o mini writes suggestions from scraped recipes found by SentenceBERT vector search in ChromaDB, and internal testers rated **93%** of meals satisfactory

## PROJECTS

**IEX Order Book & Matching Engine** | *C++17, STL, POSIX, Linux perf*

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Implemented B+ tree and LSM-tree backends for the event log and benchmarked them on identical 2M-record workloads, where the B+ tree gave **12.8x** the read throughput and 22x faster range queries at the cost of 27% more disk space
- Sustained **1.26M events/sec** end-to-end over the full replay, keeping each price level's orders in a std::list FIFO inside a std::map and a hash map from order ID to list iterator for O(1) cancels

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, C, TypeScript, JavaScript, SQL, Bash

**Cloud & DevOps:** AWS (Lambda, Kinesis, Fargate, EC2, API Gateway, EventBridge, CloudWatch), AWS CDK (infrastructure as code), Docker, GitHub Actions (CI/CD)

**Backend & Full-Stack:** REST APIs, Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

**AI/ML:** LLMs (Claude, Bedrock, GPT-4o mini), retrieval-augmented generation (RAG), fine-tuning and evaluation, PyTorch

**Practices & Tools:** data structures and algorithms, unit testing (Pytest, Jest), profiling (perf, psutil), Linux, Git, Jira, Agile